

Zomato Customer Insights & Strategic Analysis Report

Executive Summary:

This report provides a comprehensive analysis of Zomato's customer base through three distinct lenses: **Demographics, Purchasing Behavior, and Strategic Segmentation**. My analysis identifies Gen Z students as the primary volume drivers, while working professionals represent the highest value-per-order. The following pages detail the findings and provide actionable recommendations for Zomato's growth strategy.

Page 1: Customer Profile (Demographics)

Goal: Understanding "Who" is ordering.

- **Age Distribution + Gender:** The pie chart shows a skew toward the **males** while the clustered bar chart shows the restaurant specifically caters to **18–25 age group**. This segment indicates that Zomato is primarily a "youth-driven" platform.
- **Occupation vs Monthly Income:** The clustered chart reveals that the largest group is **Students with No Income**, followed by **Employees in the 25k–50k income bracket**.

Hypothesis: Zomato's primary market penetration is Students, creating a high-volume but low-immediate-margin user base.

Page 2: Purchasing Behavior

Goal: Understanding "How" and "When" they order.

- **Families + Income Level:** A scatter plot analysis confirms a positive correlation between family size and the number of items ordered. Households of **4 to 6 members** show the highest "Average Quantity per Order," making them ideal targets for bulk-meal deals.
- **Days of the Weeks:** The Line Chart shows clear peaks during **Weekends (Friday–Sunday)**. However, high-income "Post-Graduate" segments maintain a steadier order frequency throughout the weekday, likely utilizing the app for office lunches.
- **Cuisine Preferences:** Through the Stacked Bar Chart, it shows that while "North Indian" and "Chinese" are the favorites.
- **Hypotheses:**
 - Larger households optimize delivery costs by increasing the quantity of items per transaction.
 - Students drive weekend volume spikes, while professionals provide a stable weekday baseline.

Page 3: Strategic Segments

Goal: Data-driven groupings for targeted marketing.

- **Cluster Analysis (Total Revenue vs. Order Frequency)**
 - **The "VIP" Cluster:** A small group of users (Top-Right) responsible for a disproportionate share of revenue. These are **high-frequency, high-spend individuals**.
 - **The "Budget" Cluster:** Users in the Bottom-Left who only order low-cost items sporadically. They are **low frequency, low spend individuals**.
 - **The "Frequent" Cluster:** Loyal, middle-tier users **who order frequently but maintain a lower spending per order**.
- **Top 10 Restaurants for the "Post-Graduate" Segment**
 - By filtering for the most educated demographic, the preferences shift toward standard Fast-Food chains.
 - This segment prioritizes "lowest price" over quality and variety, so marketing to this group should focus on "Quality and Taste" rather than "Flash Sales."
- **Hypothesis:** High-income users treat Zomato as a premium service for curated meals while budget-conscious users treat it as a convenience for frequent, small snacks or fast food.

Methodology & Data Integrity

1. Data Preparation and Transformation

Before building the visuals, the raw Zomato datasets were cleaned and structured to ensure accuracy:

- **Data Type Standardization:** Columns such as **rating** and **rating_count** were converted from text to **Decimal** and **Whole Number** respectively to allow for mathematical calculations like Average and Sum.
- **Null Value Management:** Rows with missing restaurant names or critical user demographics were filtered out to prevent "Blank" categories from skewing the percentages in the charts.
- **Feature Engineering:** * A **"Day of Week"** column was created using DAX (FORMAT('order_date', "dddd")) to enable the weekend vs. weekday trend analysis.
 - An **"Average Quantity"** measure was calculated to bridge the gap between individual orders and family-sized purchasing behavior.

2. Preprocessing Assumptions & Clustering Logic

To generate the **Strategic Segments** on Page 3, the following logic was applied:

- Since "Order Count" (small numbers) and "Total Revenue" (large numbers) exist on different scales, Power BI's clustering algorithm **automatically normalized** the data. This prevents the high revenue values from not getting intertwined with the frequency data, ensuring both metrics carry equal weight in defining a "VIP" customer.
- The clusters were defined by assigning users to a group based on the shortest distance to an average point of that cluster. I chose **3 clusters** to create a clear hierarchy: **High Value**, **Frequent**, and **Budget**.

3. Limitations

While the dashboard provides strong insights, the following limitations must be considered:

- **Self-Reported Data Bias:** Information such as "Monthly Income" and "Educational Qualifications" is self-reported by users. There is a potential for **desirability bias**, where users may overstate their income or education levels.
- **Missing Values in "Rating":** A portion of the dataset contained "0" or "NaN" ratings. The analysis assumes these are "not yet rated" rather than "poorly rated," but this could slightly lower the perceived performance of some restaurants.
- **Sample Representativeness:** The data is heavily skewed toward the **Gen Z group (18–25 age range)**. While this accurately reflects Zomato's current core user base, the findings may not be generalized to older demographics (45+) who may have different ordering habits not fully captured in this data.
- **Geographic Constraint:** The analysis assumes a uniform delivery environment. Factors like "Delivery Distance", which significantly impact customer satisfaction, were not available in this dataset and may have helped in the analysis.

Recommendations

- **Introduce "Zomato Rewards":** Create a subscription tier that offers "Buy 3 Get 1 Free" or "Mega-Bucket" discounts specifically on weekends to capture the large family market identified in the **Purchasing Behavior** page.
- **Hyper-Targeted Ads:** Use the **Strategic Segments** page to send personalized notifications to certain groups. "Budget Users" should receive more discounts, while "High-Value Users" should receive "Gourmet Exclusive" incentives.
- **Corporate Lunch Partnerships:** Focus on the "Employee" demographic during the week by offering specialized office-delivery bundles.